

Geodesics and hyperbolicity for half-plane models of taxicab hyperbolic space

Luka Lawrence, Kevin Lin, Alina Potapyyev

Seattle University

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Plan

- Taxicab geometry

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- Hyperbolic space

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 - Half-plane model

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- Merging taxicab geometry with hyperbolic geometry

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- Next steps

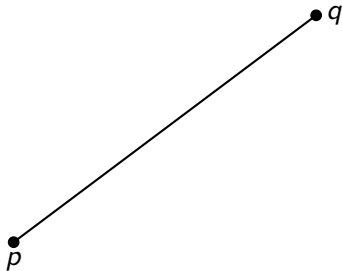
Taxicab geometry

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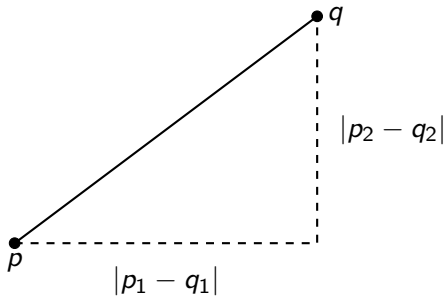
p

q

Taxicab geometry

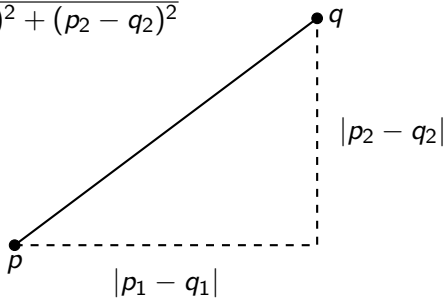


Taxicab geometry



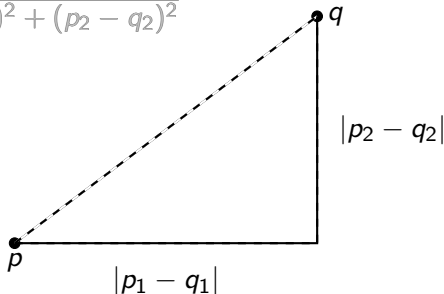
Taxicab geometry

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$$d_T(p, q) = |p_1 - q_1| + |p_2 - q_2|$$

Hyperbolic geometry

Half-plane model

Merging taxicab geometry with hyperbolic geometry

Use Alina's slide here

- translations parallel to ℓ

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- dilations centered at a point on ℓ

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- reflections across lines perpendicular to ℓ (only if $m = 0$ or 1)

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 - dilations centered at a point on ℓ
 - reflections across lines perpendicular to ℓ (only if $m = 0$ or 1)
 - no others (not obvious, and technical to prove)
- ☐ add pictures?

Special curves

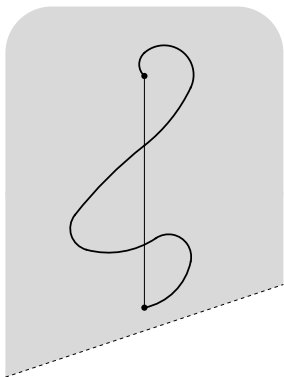
- $\sigma_{p,q}$ = line segment between p and q

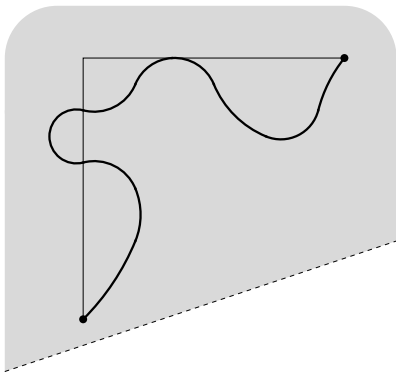
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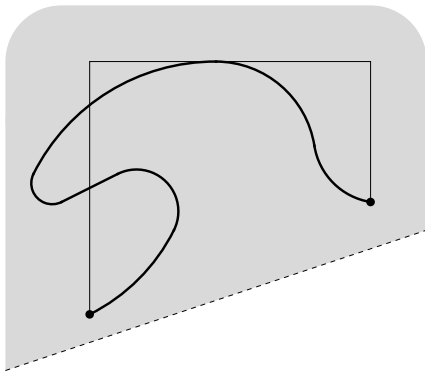
- $\sigma_{p,q}$ = line segment between p and q
- $\lambda_{p,q} =$
- $\eta_{p,q} =$
- ☐ finish descriptions
- ☐ need to decide about notation for η ... How to account for variable height? Or can we get away with just the optimal value?

Finding length minimizing curves

- pictures of various cases, producing a general η

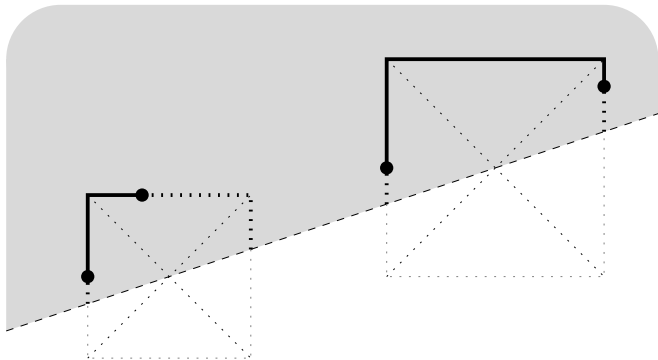






Sample estimate

Geodesics



- ☐ distance function
- ☐ connection between terms in formula and parts of η
- ☐ fun fact

Given a distance function d , the Gromov product is

$$G(x, y; z) = \frac{1}{2}[d(x, z) + d(y, z) - d(x, y)]$$

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A space is δ -hyperbolic, in the sense of Gromov if

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Rewriting this inequality, we define the hyperbolicity function

$$h(x, y, z, w) = \min\{G(x, z; w), G(y, z; w)\} - G(x, y; w)$$

and aim to find an upper bound, δ , for this function.

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and the Gromov product:

$$G(x, y; z) = \frac{1}{2} [d(x, z) + d(y, z) - d(x, y)]$$

$$\begin{aligned} & \frac{1}{2} \ln \left(\frac{\Omega(x, z) \Omega(y, z)}{\Omega(x, y) (z_2 - mz_1)^2} \right) - \frac{1}{2m} \ln \left(\frac{1+m}{1-m} \right) \\ & \leq \\ & G(x, y; z) \\ & \leq \\ & \frac{1}{2} \ln \left(\frac{\Omega(x, z) \Omega(y, z)}{\Omega(x, y) (z_2 - mz_1)^2} \right) + \frac{1}{m} \ln \left(\frac{1+m}{1-m} \right) \end{aligned}$$

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where $\Omega(p, q) = (\mu(p, q) - mp_1)(\mu(p, q) - mq_1)$.

With these bounds, the hyperbolicity function

$$h(x, y, z, w) = \min\{G(x, z; w), G(y, z; w)\} - G(x, y; w)$$

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Note symmetries: $x \leftrightarrow y$, $z \leftrightarrow w$, and $(x, y) \leftrightarrow (z, w)$.

Combining results of various cases, the hyperbolicity function is bounded by

$$\delta = \ln(4) + \frac{4m+3}{2m} \ln \left(\frac{1+m}{1-m} \right).$$

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- limit as $m \rightarrow 0^+$ agrees with our bound for $m = 0$
- diverges as $m \rightarrow \infty$, but $\mathbb{H}_1$ is hyperbolic

Next steps

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- Explore quotient spaces

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- Explore quotient spaces
- Explore L^p -hyperbolic spaces

Next steps

- Explore quotient spaces
- Explore L^p -hyperbolic spaces
- Explore higher dimensions

THANK YOU!
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Questions?